

Preliminaries

- Correctness and Loop Invariants
- Case Study: Sorting
- Case Study: Binary Search
- Case Study: KMP String Matching (X.ref week 7)

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Algorithm v.s. Correct Algorithm

- An algorithm passes: all provided test cases, and 1,000,000 random tests. Is it correct?
 - Just vote: **Yes / No / Not sure.**
- Algorithm vs. Correct Algorithm:

Algorithm

A finite, well-defined, executable procedure.

Correctness

A property we prove *about* an algorithm. An algorithm may be incorrect, but it is still an algorithm.

- **Definition.** An algorithm is *correct* if, for every valid input, it terminates and produces an output that satisfies the problem specification.
- This definition has three essential components:
 - universality (for all inputs),
 - Termination
 - specification satisfaction.

Claim (incorrect)

“The algorithm works because I tested it on many random inputs.”

Failure analysis:

- The input space is typically huge (often effectively infinite)
- Testing covers only finitely many cases
- Bugs often appear only in corner cases (boundaries, duplicates, off-by-one)

What Proofs Protect You From

- Passing all tests but still being wrong
- Infinite loops you didn't anticipate
- Boundary cases you forgot
- “Seems correct” intuition

Correctness of Iterative Algorithms

- Most algorithms in this course are: iterative, state-based (arrays, indices, pointers), gradually transforming the input into the output.
 - We must therefore reason about intermediate states, not just the final result.
- Central Question:
 - During the execution of a loop, what is guaranteed to be correct so far?

Answer

A **loop invariant** captures these guarantees at iteration boundaries.

Example: Proving the Correctness of FindMax

Algorithm FindMax(A)

Input: An array A with n elements

Output: The maximum value in A

```
1. max_so_far ← A[1]
2. for i ← 2 to length(A) do
3.     if A[i] > max_so_far then
4.         max_so_far ← A[i]
5.     end if
6. end for
7. return max_so_far
```

Proof of Correctness

Goal: Showing that max_so_far stores the maximum value of A.

1. Before the Loop Starts (Initialization)

- Before entering the loop, max_so_far = A[1]. Since we've only seen one element, this is correct.

2. During Each Iteration (Maintenance)

- If $A[i] > \text{max_so_far}$, we update $\text{max_so_far} = A[i]$, ensuring it holds the maximum seen so far.
- Otherwise, max_so_far remains unchanged, which is still correct.

3. After the Loop Ends (Termination)

- At the end, max_so_far contains the maximum of all A[1] to A[n].

- A **loop invariant** is a property or condition that holds true before and after each iteration of a loop.
- Purpose:
 - To show that an algorithm maintains a specific condition throughout its execution.
 - To help prove that the algorithm works correctly (via **initialization, maintenance, and termination**).
- The way that we prove FindMax correct is Loop Invariant.
 - Loop invariant: max_so_far holds the maximum value among A[1:i].

Note

The invariant is required to hold at iteration boundaries (start of iterations), not during the loop body.

- **Initialization**: Show that the invariant holds **before** the first iteration (base case).
- **Maintenance**: Assuming the invariant holds at the beginning of **any** iteration, **prove that it still holds** after executing the loop body.
- **Termination**: When the loop terminates, the invariant (plus the loop's exit condition) gives a useful property that helps prove the algorithm's correctness.

Warning

Omitting any component invalidates the proof.

*FYI. The three steps is introduced in **CLRS**. In other books, you may find Establishment (i.e. Initialization), Preservation (i.e. Maintenance), Postcondition and Termination: Postcondition ensures the final goal is achieved if the loop stops; Termination guarantees that the loop will stop.*

A Subtle but Common Failure

Claimed invariant (incorrect)

“The invariant holds because it is true after one iteration.”

Failure analysis:

- This only argues maintenance
- Initialization is never established
- No anchor to the execution start

Bottom line

Maintenance alone is never sufficient.

How to Interpret an Invariant

Read it like this

“When the loop index is i , this part of the algorithm’s goal has already been achieved.”

It describes:

- What is already correct
- What remains to be done
- How correctness progresses with i

Not all invariants are equally useful. An invariant can be:

- Too weak — easy to maintain, but insufficient to imply correctness
- Too strong — implies correctness, but hard or impossible to maintain
- Just strong enough

Weak Invariant Example

Weak invariant (sorting)

“Some elements are already in the correct position.”

Why this fails:

- Does not specify which elements
- Does not depend on i
- Cannot imply full sortedness at termination

Overly Strong Invariant Example

Overly strong invariant

“At the start of iteration i , the entire array is sorted.”

Why this fails:

- False during early iterations
- Cannot be established during initialization
- Breaks the maintenance step

The “Just-Right” Invariant

A good invariant typically:

- Refers explicitly to loop index i
- Fixes a prefix, suffix, or subset
- Grows monotonically as i increases

Such invariants are:

- Provable
- Maintainable
- Sufficient for correctness

How to find invariant

Treat invariant design as a mechanical procedure:

- Write the final correctness condition (postcondition)
- Identify what is not yet guaranteed early in the loop
- Remove that part
- Index what remains by the loop variable(s)

Typical failure mode:

- Start from intuition
- Jump directly to the final result
- Skip intermediate guarantees

Takeaway

This is an algorithm: you can apply it step-by-step on exams.

Standard Loop Invariant Pattern (Array Iteration)

Template

“At the start of iteration i , the substructure [explicit range depending on i] satisfies [precise property].”

Rule of thumb

Any invariant not fitting this template should be treated with suspicion.

Case study: Sorting

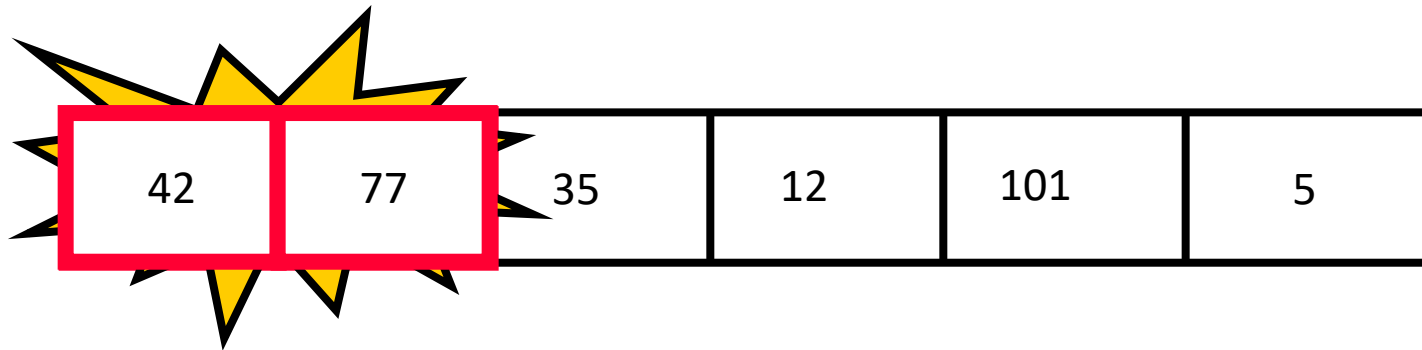
Bubble Sort: "Bubbling Up" the Largest Element

- Traverse a collection of elements
- Move from the front to the end
- “Bubble” the **largest value** to the end using **pair-wise comparisons and swapping**

77	42	35	12	101	5
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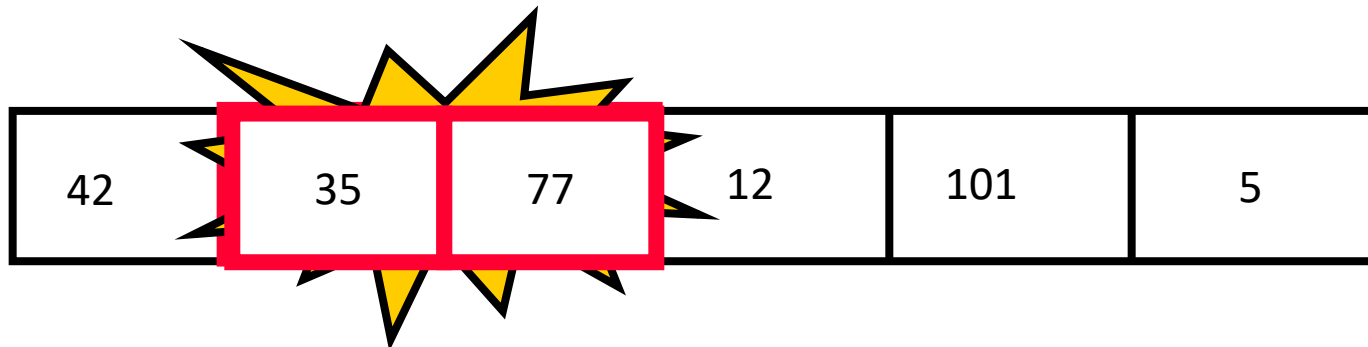
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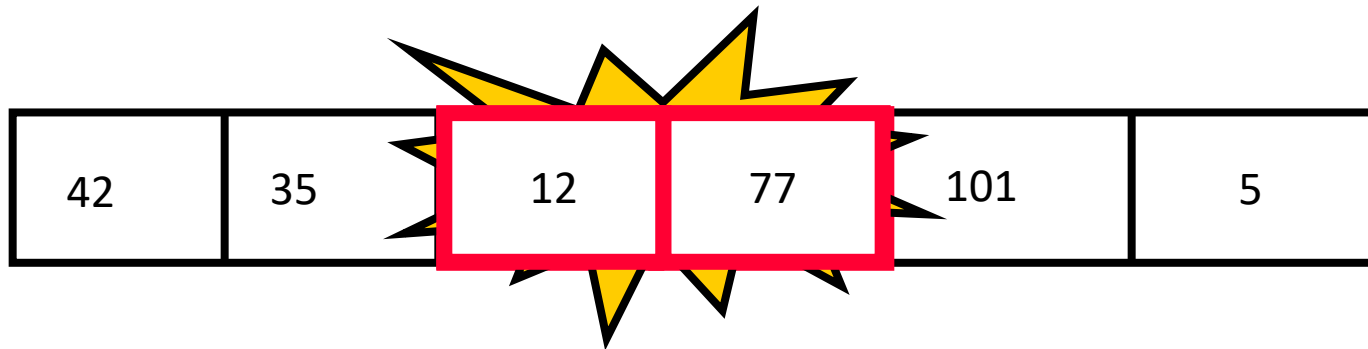
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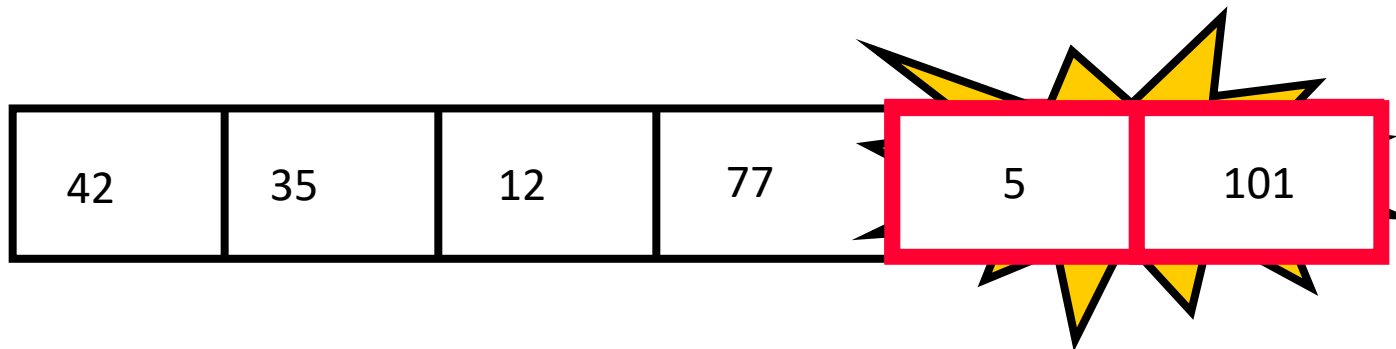
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No need to swap

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42	35	12	77	5	101
----	----	----	----	---	-----

Largest value correctly placed

The “Bubble Up” Algorithm

```
index ← 1
last_compare_at ← n - 1

while index < last_compare_at+1
  if(A[index] > A[index + 1]) then
    Swap(A[index], A[index + 1])
  end if
  index ← index + 1
end while
```

- Notice that only the largest value is correctly placed
- All other values are still out of order
- So we need to repeat this process

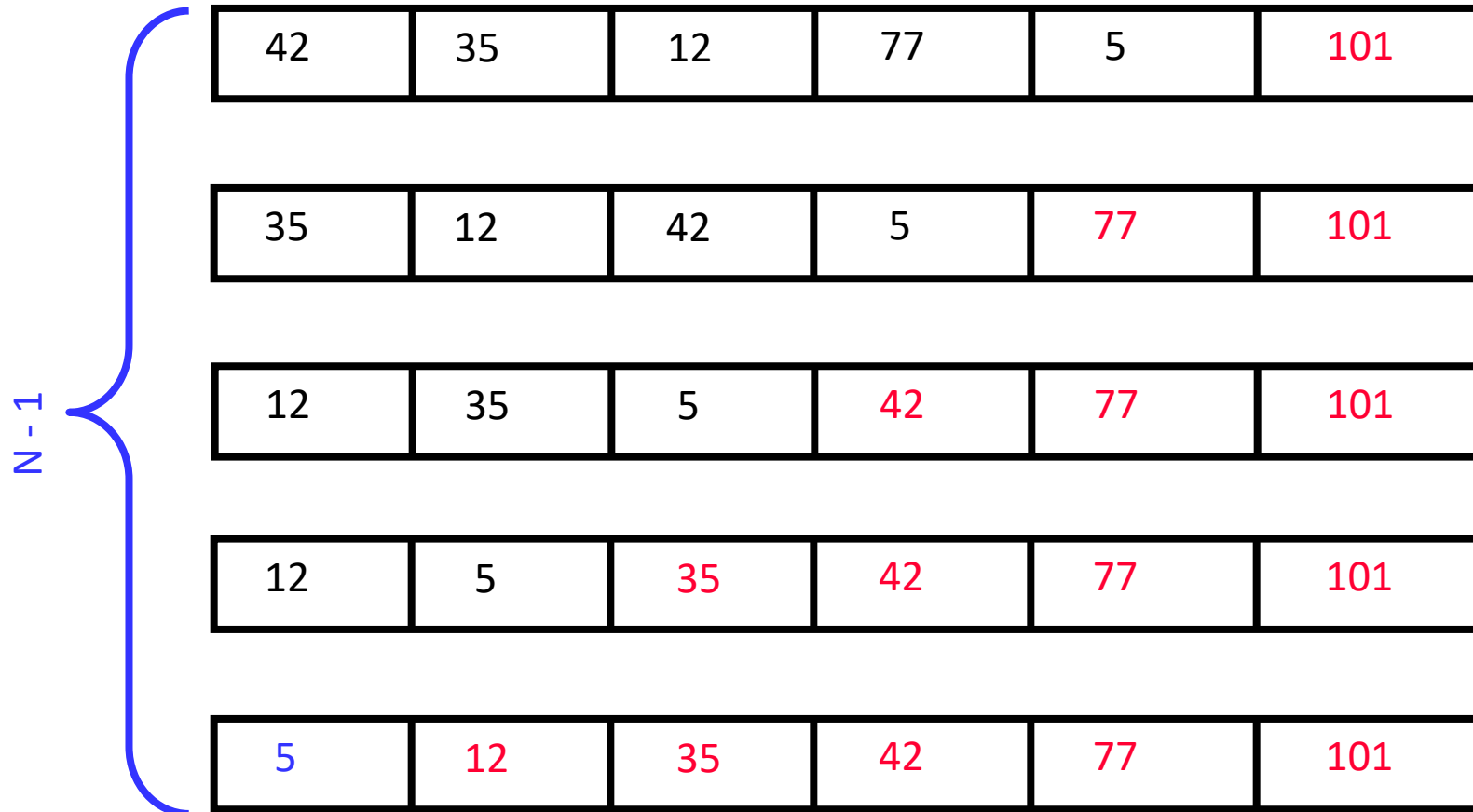
42	35	12	77	5	101
----	----	----	----	---	-----

Largest value correctly placed

Repeat “Bubble Up” How Many Times?

- If we have N elements ...
- And if each time we bubble an element, we place it in its correct location ...
- Then we repeat the “bubble up” process $N - 1$ times
- This guarantees we’ll correctly place all N elements

“Bubbling” All the Elements



Reducing the Number of Comparisons

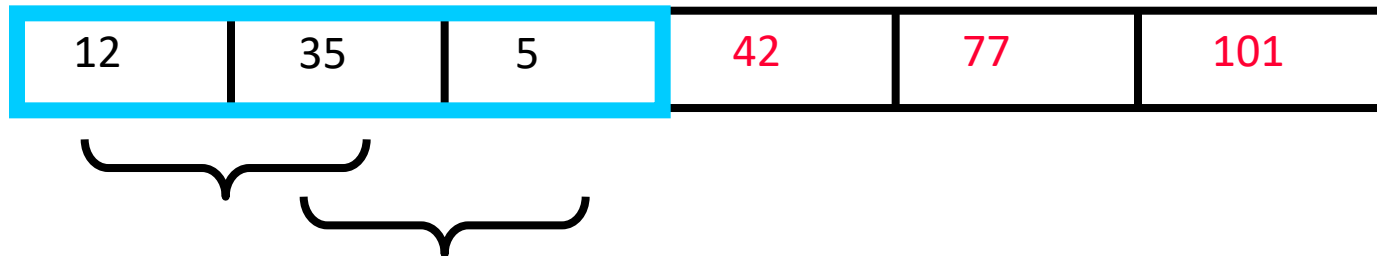
77	42	35	12	101	5
42	35	12	77	5	101
35	12	42	5	77	101
12	35	5	42	77	101
12	5	35	42	77	101

Reducing the Number of Comparisons

- On the i^{th} “bubble up”, we only need to do $\text{MAX} - i$ comparisons

For example:

- This is the 4th “bubble up”
- MAX is 6
- Thus we have 2 comparisons to do



Putting It All Together

```
algorithm Bubblesort(A)
  to_do, index isoftype Num
  to_do ← N - 1

  while to_do > 0:
    index ← 1
    while index < to_do + 1
      if(A[index] > A[index + 1]) then
        Swap(A[index], A[index + 1])
      endif
      index ← index + 1
    end while
    to_do ← to_do - 1
  end while
```

Inner loop

Outer loop

Already Sorted Collections?

- What if the collection was already sorted?
- What if only a few elements were out of place and after a couple of “bubble ups,” the collection was sorted?
- We want to be able to **detect this and “stop early”!**

5	12	35	42	77	101
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Using a Boolean “Flag”

- We can use a boolean variable to determine if any swapping occurred during the “bubble up”
- If no swapping occurred, then we know that the collection is already sorted!
- This boolean “flag” needs to be reset after each “bubble up”

```
algorithm Bubblesort(A)
  to_do: index isoftype Num
  did_swap: Boolean
  to_do  $\leftarrow$  N - 1
  did_swap  $\leftarrow$  true

  while (to_do > 0):
    index  $\leftarrow$  1
    did_swap  $\leftarrow$  false
    while index < to_do + 1
      if(A[index] > A[index + 1]) then
        Swap(A[index], A[index + 1])
        did_swap  $\leftarrow$  true
      endif
      index  $\leftarrow$  index + 1
    end while
    to_do  $\leftarrow$  to_do - 1
    if did_swap = false: break
  end while
```

Which Loop Should We Analyze?

- Bubble sort has an inner loop and an outer loop
- The inner loop performs local comparisons
- The outer loop makes global progress
- Correctness is proved using the outer loop invariant

Think First: What Is Already Correct?

- Suppose k outer-loop iterations have completed
- Which elements will never move again?
- Which part of the array is already sorted?

Proving Bubble Sort (w.r.t. Outer Loop)

```
algorithm Bubblesort(A)
  to_do: index isotype Num
  did_swap: Boolean
  to_do ← N - 1
  did_swap ← true

  while (to_do > 0):
    index ← 1
    did_swap ← false
    while index < to_do + 1
      if(A[index] > A[index + 1]) then
        Swap(A[index], A[index + 1])
        did_swap ← true
      endif
      index ← index + 1
    end while
    to_do ← to_do - 1
    if did_swap = false: break
  end while
```

Proof of Correctness

Loop Invariant: $A[to_do+1:n]$ is a sorted permutation of largest elements in final positions

1. Initialization

- Before the first iteration, $to_do = n - 1$; The suffix $A[n+1..n]$ is empty; An empty suffix is trivially sorted

2. Maintenance

- Inner loop moves the maximum of $A[1..to_do+1]$ to position to_do+1 ; Then to_do is decremented → Sorted suffix grows by one element

3. Termination

- Loop ends when $to_do = 0$ or no swaps occur. Invariant implies entire array $A[1..n]$ is sorted. Correctness follows directly

Algorithm InsertionSort(A)

Input: An array A with n elements

Output: A sorted in non-decreasing order

```
1. for i ← 2 to length(A) do
2.     key ← A[i]
3.     j ← i - 1
4.     while j > 0 and A[j] > key do
5.         A[j + 1] ← A[j]
6.         j ← j - 1
7.     end while
8.     A[j + 1] ← key
9. end for
10. return A
```

Proof of Correctness

Loop Invariant: $A[1:i-1]$ is a sorted permutation of elements in the original $A[1:i-1]$

1. Initialization (Note: $i \leftarrow 2$ is not part of the loop)

- When $i=2$, $A[1:1]$ contains one value $\rightarrow$ sorted.

2. Maintenance

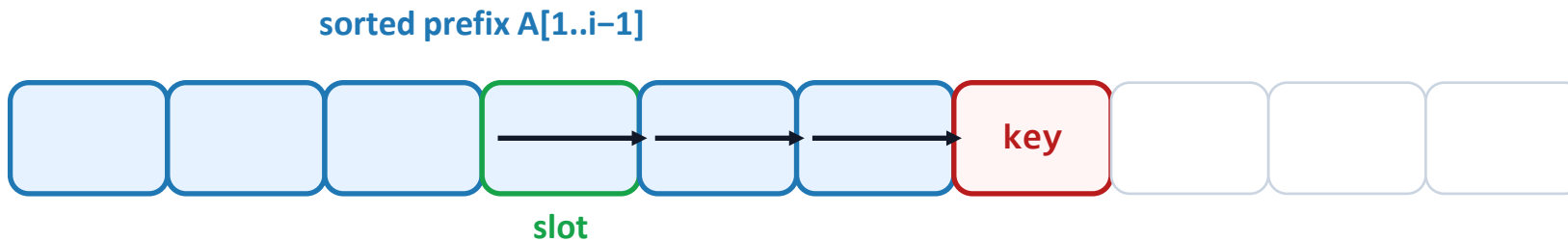
- Within the loop-body, $A[1:i-1]$ is assumed to be sorted. We move $A[i-1]$, $A[i-2]$, ... toward the right, until we find a position for $A[i]$. Once $A[i]$ is inserted, $A[1:i]$ remains sorted.

3. Termination

- As i goes from 2 to $\text{length}(A)$, and the loop body does not modify i . The loop will terminate. By termination, $i=\text{length}(A)$ means $A[1:\text{length}(A)]$ is sorted.

Role of the Inner Loop in Insertion Sort

- Shifts elements larger than key one position to the right
- Creates exactly one “slot” for key



Crucially:

- Preserves the relative order of the shifted block
- Does not disturb elements $\leq$ key

For a fully formal proof, an inner-loop invariant can help, but the outer-loop invariant is the main driver.

Common Trap (Think 30s)

False statement

“The inner while loop sorts the array, so insertion sort is correct.”

Your job

Which part of the array does the inner loop affect, and why is that not enough?

Rule of thumb

Correctness is proved by an invariant tied to the outer loop.

Case Study: Binary Search

Binary Search

Input

A sorted array $A[1..n]$ (nondecreasing) and a target value x .

Goal

Return whether there exists an index k such that $A[k] = x$.

Binary search exploits sorted order to reduce the candidate region.

Find an element in a sorted array:

1. Check middle element.
2. Recursively search 1 subarray.

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Example: Find 9

3 5 7 8 9 12 15

Find an element in a sorted array:

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Example: Find **9**

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Find an element in a sorted array:

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A Nearly Correct Binary Search

```
1 low = 1
2 high = n
3 while low < high:
4     mid = low + (high - low) // 2
5     if A[mid] < x:
6         low = mid
7     else:
8         high = mid
.....
```

What is wrong with this algorithm?

A Concrete Failure Case

Let

$A = [1, 2], \quad x = 2$

Execution trace:

- $\text{low} = 0, \text{high} = 1$
- $\text{mid} = 0$
- $A[\text{mid}] < x \Rightarrow \text{low} \leftarrow \text{mid} = 0$

What happens?

The loop never makes progress.

How to Fix the Bug

Invariant we want

If x exists in A , then it lies in $A[\text{low}..\text{high}]$.

Question: Which update guarantees progress when $A[\text{mid}] < x$?

- $\text{low} \leftarrow \text{mid}$
- $\text{low} \leftarrow \text{mid} + 1$

Explain

Why must at least one index be removed each iteration?

1) Preserve the invariant

If x exists, it remains in $A[\text{low}..\text{high}]$.

2) Strictly shrink the interval (variant)

At least one index is eliminated each iteration (so $\text{high}-\text{low}$ decreases).

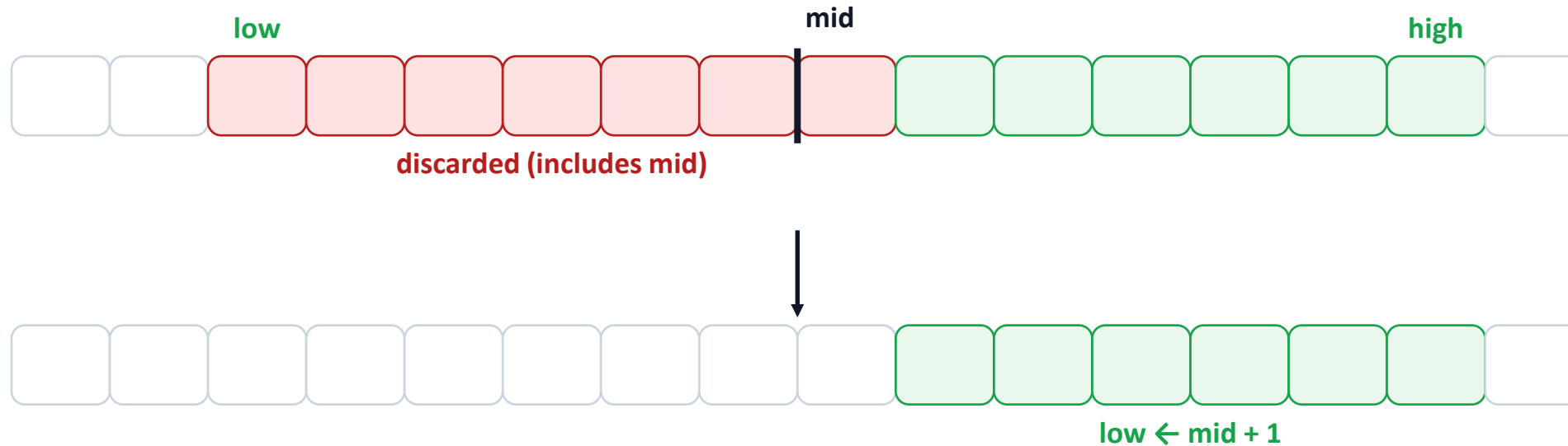
Invariant $\Rightarrow$ partial correctness; shrinking $\Rightarrow$ termination; together $\Rightarrow$ correctness.

Correct Binary Search Algorithm

```
1 low = 1
2 high = n
3 while low <= high:
4     mid = low + (high - low) // 2
5     if A[mid] == x:
6         return true
7     else if A[mid] < x:
8         low = mid + 1
9     else:
10         high = mid - 1
11 return false
```

Binary Search: The Interval Must Shrink

Invariant: if x is present, it lies in $A[\text{low}..\text{high}]$.



We eliminate at least one index each iteration $\Rightarrow$ termination.

Invariant

At the start of each iteration, if the target value x exists in A , then it must lie in the subarray $A[\text{low}..\text{high}]$.

Case Study: KMP String Match

Scope: What We Do (and Do Not) Cover

- We use string matching only as an example to practice loop-invariant reasoning, especially two-index invariants involving (i, j) .
- We do not teach the full KMP algorithm today.
 - The full algorithm will be introduced in Week 7.

 Input: Text $T[1..n]$ and pattern $P[1..m]$.

 Output: Decide whether P appears as a contiguous substring of T .

Naïve approach: try alignments and compare left-to-right. When a mismatch happens, it feels natural to “**shift** the pattern and try again”.

How many comparisons are required? – $O(___)$

The key correctness question (not an efficiency question)

How do we know a shift does not skip a valid match?

Why Correctness Is Non-Trivial Here

- During matching, we repeatedly build a partial match and then break it by a mismatch.
- A careless shift can discard positions that still could be the start of a match.
- Testing cannot cover all texts/patterns; correctness must be a logical guarantee.

What do we need?

A statement that captures exactly what we know is already correct at any time. → *two-index loop invariant for this example*

State Definition: What Do i and j Mean?

- To state an invariant, we must define the state precisely:
 - i: current text position (the next text character to be processed/compared)
 - j: the number of pattern characters currently matched
- Meaning of “j matched”: The prefix $P[1..j]$ matches the suffix of already processed text:

$$P[1..j] = T[i-j..i-1].$$

(If $j=0$, the matched prefix is empty and the statement is trivially true.)

- Loop Invariant statement: At the start of every iteration boundary (with current (i, j)), $P[1..j]$ matches a suffix of $T[1..i-1]$.
 - The algorithm has already certified a length- j match ending at position $i-1$ in T .
 - All future updates must preserve this certified fact.

- Suppose we are about to compare $T[i]$ with $P[j+1]$, and they match.
We then,
 - Increase i by 1.
 - Increase j by 1.
- Matching grows the invariant $\rightarrow$ The invariant still holds.

Mismatch: What Must Any “Jump” Preserve?

- Suppose we are about to compare $T[i]$ with $P[j+1]$, and they mismatch.
- Constraints we impose (the KMP philosophy, X.ref week 7):
 - We do not move the text pointer backward (i never decreases).
 - So the only freedom is to reduce j (try a shorter matched prefix).
 - But we are not allowed to reduce j arbitrarily.
 - The new value j' must keep the invariant true. $P[1..j'] = T[i - j'..i - 1]$.

Key message

A jump is safe iff it preserves the invariant. Any update violating the invariant is incorrect.

Do we skip a better alignment?

- When mismatch: There may exist multiple values of j that satisfy the invariant. The invariant does **not** claim:
 - that the current j is the largest
 - or the only possible one
 - The invariant guarantees **non-exclusion**, not optimality.(*Will introduce how to choose j which satisfies this invariant in week 7.)

Loop Invariant:

At the beginning of each iteration (with current indices i, j), $P[1..j] = T[i-j .. i-1]$. That is, the prefix $P[1..j]$ matches a suffix of the processed text $T[1..i-1]$.

1. Initialization

Initially, $i = 1$ and $j = 0$. Both sides represent the empty string, so the invariant holds.

2. Maintenance

- If $T[i] = P[j+1]$, both i and j increase by one, extending the certified match.
- If $T[i] \neq P[j+1]$, j is reduced while keeping i fixed, but only to values that preserve the invariant.

3. Termination

If $j = m$, then by the invariant $P[1..m] = T[i-m .. i-1]$, so a correct match is found.

If the text ends without reaching $j = m$, then no match exists.

All our examples share the same structure: an invariant certifies a part of progress.

Algorithm	Certified object (invariant)
Max scan	$\text{max} = \text{max}(A[1..i-1])$ (certified prefix)
Insertion sort	$A[1..i-1]$ is sorted (certified sorted prefix)
Bubble sort	$A[\text{to_do}+1..n]$ sorted (certified sorted suffix)
String matching	$P[1..j] = T[i-j..i-1]$ (certified match length)